

School of Post Graduate Studies

Master of Information Technology (MIT)

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2026/2027 Academic Session
Continuous Assessment

MIT 8333 - Advanced Web Application Development
(Virtual lab)

Submission Guidelines

You must submit in one PDF:

1. All the code base
2. Screenshots of the output
3. Project report

Project Scenario

The university currently receives maintenance complaints and service requests manually through phone calls, paper forms, WhatsApp messages, and office visits. This process causes delays, missing records, poor tracking, and lack of accountability. Students and staff need a digital platform where they can submit service requests such as faulty electricity, damaged furniture, leaking pipes, internet problems, classroom equipment issues, and hostel maintenance complaints. Maintenance officers should be able to view assigned requests, update progress, and mark jobs as completed. Administrators should be able to manage users, assign tasks, monitor request status, and generate reports.

You are required to develop, test, and deploy a full-stack web application that solves this problem.

Project Requirements

A. Frontend Development

The system should include:

1. User registration and login pages.
2. Role-based dashboard for:
 - Student/Staff
 - Maintenance Officer
 - Administrator

3. Service request submission form.
4. Request tracking page.
5. Admin request management interface.
6. Good navigation, layout, form validation, and user feedback messages.

B. Backend/API Development

The backend should provide:

1. User authentication and authorization.
2. RESTful or GraphQL APIs.
3. CRUD operations for service requests.
4. Role-based access control.
5. Request assignment to maintenance officers.
6. Error handling and validation.
7. Secure password handling and session management.

C. Database Integration

The application should use a relational or NoSQL database.

Minimum database entities:

1. Users
2. Roles
3. Service Requests
4. Request Categories
5. Assignments
6. Status Updates/Logs

D. Advanced Web Application Features

Implement at least four of the following:

1. JWT/session-based authentication.
2. Role-based access control.
3. File/image upload for evidence of fault.
4. Email or in-app notifications.
5. Search, filter, and pagination.
6. Real-time status updates.
7. Audit trail/activity log.
8. API documentation using Swagger/Postman.
9. Data export to PDF or CSV.

E. Testing and Deployment

You must:

1. Test major frontend components.
2. Test backend API endpoints.
3. Deploy the application online. Ensure the deployed version connects correctly to the database.

F. Technical Documentation

The project report should include:

1. Introduction and problem statement.
2. System objectives.
3. Requirement analysis.
4. Frontend technologies used
5. Backend technologies used
6. The database used and the types of relationships it supports.
7. API documentation.
8. Screenshots of major interfaces.
9. Testing evidence.
10. Deployment information.
11. Challenges encountered and solutions.
12. Conclusion.